# Surya Ravi

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## About Me

Engineer with cross-domain knowledge spanning cybersecurity, computer science, AI and system design. Experienced in building event-driven automation flows and AI systems for threat exposure management, SOAR automation and vulnerability management, and exploit-verified security tooling with static analysis and LLM agents.

## Experience

**Orbia** Jul. 2026 – Present
*Threat Exposure Management Analyst* *Pune, Maharashtra*

- Built an event-driven **zero-day and N-day** pipeline in **Tines** using **webhooks** and **HTTP Request** actions that ingests critical vulnerabilities from **Assetnote** and **CrowdStrike** on internet-facing domains, subdomains, IPs and cloud assets, alerts owners through **Outlook** and **Microsoft Teams**, and raises **ServiceNow** tickets for the responsible assignment group, cutting time to remediation.
- Integrated **CrowdStrike Falcon Fusion SOAR** with **Tines** to detect cloud misconfiguration drift, alerting the team on critical **Indicators of Misconfiguration (IOMs)** and documenting and tracking remediation in **ServiceNow**; developed **5** Falcon Fusion workflows for automated response and remediation.
- Engineered a **false positive reduction engine** with the **Netskope** team that re-filters malicious alerts so only true positives reach analysts, allowlisting validated activity per business domain.
- Built firewall control-failure monitoring integrating **Netskope**, **FortiGate** and **Palo Alto Panorama** to detect missing **GRE** and **IPsec** tunnels to Netskope PoPs by geographic region and raise **ServiceNow** tickets.
- Automated monthly **Proofpoint** and **Splunk** email-threat reporting on phishing, impostor, malicious, duplicate and other mail.
- Built a **GitHub** pull-request automation with a stateful **LangGraph** agent that parses PR diffs, flags malicious code for review and stores reviewed verdicts in long-term memory to keep improving.
- Built databases and dashboards in **Tines**, backed by a relational **SQL** database, to retain automation data for future reference and visualise daily and monthly metrics for **compliance and audit**.
- Own quarterly **SAP** and **internet-facing** penetration tests on **Intigriti** end to end, from cleaning scope and clearing backlog vulnerabilities to closing reports with the **GRC** team, performing retests to validate patches and variant analysis, and reporting to business-group owners and stakeholders.
- Executed **2** white-box pentests on internal web apps and validated remediation by retesting **8** internal applications.
- Mentored interns and juniors on building and handling automations; wrote **runbooks** and **SOPs** for the workflows.

**Dura-Line (Orbia)** Dec. 2025 – Jun. 2026
*Security Analyst Intern* *Remote*

- Designed, developed and maintained end-to-end automations in **Tines SOAR** covering threat monitoring, alert triage, vulnerability management and false positive reclassification, built on **REST APIs**, **JSON** and **Python**.
- Built Tines automations across identity and authentication schemes, including **Basic auth**, **API keys**, **JWT** and **OAuth 2.0**, to integrate securely with security platforms.
- Triaged bug bounty and penetration testing reports in **Intigriti**, validating vulnerabilities, proof-of-concepts and remediation recommendations.
- Ramped up quickly on the enterprise security stack, moving from onboarding to building production automations.
- Took ownership of **Assetnote** and **Tenable**, continuously monitoring scanner sensor and agent health, running weekly vulnerability scans and documenting findings.
- Owned business groups' remediation tracking with stakeholders and presented progress in review meetings.

**Alice** 2026
*AI Red Team Researcher – Freelance* *Remote*

- Conducted **AI red-team** and adversarial security testing using Alice, developing attack scenarios to evaluate model behaviour, prompt-injection resistance, guardrails and security boundaries.
- Built and iterated on **automated security-testing harnesses** for adversarial prompt execution, response capture, reproducibility and regression testing across attack scenarios.
- Applied **prompt injection**, **instruction-hierarchy manipulation**, model steering and output-format coercion techniques to probe LLM applications for jailbreaks, system-prompt leakage and sensitive data exposure.
- Mapped findings to the **OWASP Top 10 for LLM Applications** and reported them with severity ratings, reproduction steps and mitigation guidance for model and product teams.

**Hackup Technology** Aug. 2025 – Nov. 2025
*Cybersecurity Analyst Intern* *Coimbatore, Tamil Nadu*

- Tested web applications with **Burp Suite**, identifying and validating security weaknesses.
- Analysed network traffic with **Wireshark**, examining protocols and packet flows to detect anomalous activity.
- Delivered technical seminars at **3** colleges on **Linux** fundamentals, the command line and cybersecurity concepts.
- Practised exploitation, validation and remediation of OWASP-aligned vulnerabilities through **PortSwigger Web Security Academy**, including SQLi, NoSQLi, IDOR, XSS, API flaws and authentication and access control issues.

## Education

**Kingston University** — Sep. 2023 – Sep. 2024
*Master's Degree in Computer Animation (School of Computer Science & Mathematics)* — *London, United Kingdom*
Award: Distinction

**Anna University (Kumaraguru College of Technology)** — Aug. 2019 – May 2023
*Bachelor's Degree in Mechatronics & Automation Engineering* — *Chennai, Tamil Nadu*
GPA: 8.8

**PVV Higher Secondary School** — Apr. 2017 – Apr. 2019
HSC: 91.4% & SSLC: 94.8% — *Coimbatore, Tamil Nadu*

## Projects

**ProofTrace** – Exploitability-first PR security review agent
- Uses **tree-sitter** to build a source-to-sink reachability graph of the changed code; a **LangGraph** investigator running on a local **Ollama** model reasons over it, and a **Docker** validator replays the exploit live before anything is reported; **Temporal** adds retries and fans out runs to measure cost-to-recall across models. Proves a planted BOLA (CWE-639) while clearing a same-shape safe endpoint.

**checkpoint-zero** – Crash-recovery benchmark for agent workflows
- Runs the same security review job as a plain Python script, **LangGraph** with SQLite checkpoints and **Temporal**, then kills each with `kill -9`; per-node checkpoints lose 0 finished reviews, and Temporal with a 2 s heartbeat recovers automatically in 2.0 s versus 19.5 s without.

**PPx** – Prompt Pattern Index
- LangGraph pipeline that classifies prompts across 4 prompt-injection pattern families (prompt extraction, instruction override, data exposure, malicious patterns) and produces a risk score, severity rating and security report.

**Attack Surface Mapper** – Codebase attack surface discovery
- Application-understanding engine that maps a codebase in 4 discovery stages (framework, routes, authentication, external services) before vulnerability hunting; surfaced 37 routes, 4 admin routes and 2 external services on a sample Express app.

## Writeups

- *Alice LLM Security* – LLM-focused security research through prompt injection, instruction-hierarchy manipulation, adversarial prompting, model steering and output-format coercion while solving an AI security CTF. Awarded Best Technical Write-up. Read
- *QKD* – Research on Quantum Key Distribution and its cryptographic foundations, exploring quantum principles, secure key exchange, eavesdropping detection and the role of QKD in post-quantum cybersecurity. Read

## Skills

**Primary Skills**: SOAR, Incident Response, MITRE ATT&CK, Threat Intelligence, Vulnerability Research & Variant Analysis, API Security, OWASP Top 10, Cryptography, Vulnerability Management, Python, C
**Secondary Skills**: Bash, Linux, Penetration Testing, Quantum Cryptography, Tree-sitter, LangGraph, CI/CD, LLM Security, Adversarial Testing, Threat Modelling, Docker, Kubernetes, Relational and Non-relational Databases
**Certifications**: GCPC, CrowdStrike Falcon Fusion SOAR, Tines Certified Professional, Postman API Developer Associate and LangGraph Foundations
**Soft Skills**: Problem-Solving, Fast-Paced Learning and Cross-Functional Collaboration
**Hobbies**: Playing guitar and chess, solving Rubik's Cube & Sudoku

## Honours & Awards

| *Name* | *Placing* | *Type* | *Scope* | *Awarder* | *Year* |
|---|---|---|---|---|---|
| Dragos OT CTF | 1st Place | OT Security CTF | Global | Dragos | 2026 |
| Alice in Wonderland LLM CTF | Top 5 | AI Security CTF | Global | Alice | 2026 |
| Alice in Wonderland LLM CTF | Best Technical Write-up | AI Security CTF | Global | Alice | 2026 |
| Kriya CTF (Jeopardy) | 1st Place | Security CTF | IN | PSG Institute of Technology | 2020 |
| IISF Hackathon | 1st Place | Hackathon | Global | Ministry of Sci. & Tech. | 2020 |
| Robo Sumo | 1st Place | Robotics | IN | PSG Institute of Technology | 2021 |
| Outstanding Student Award | Academic & Inter-College Event Excellence | Academic Award | IN | Anna University | 2023 |
| Certificate of Recognition | Academic Merit | Academic Award | IN | Anna University | 2023 |
| Mahatma Gandhi Merit Scholarship | Top 10% of Institution | Scholarship | IN | Anna University | 2019, 2020 |

## Declaration

I, Surya Ravi, hereby declare that the information furnished above is true, complete and accurate to the best of my knowledge and belief, and that the experience, projects and achievements listed are my own work and can be verified on request.